



INDIAN INSTITUTE OF TECHNOLOGY MADRAS

Department of Electrical Engineering

Self-Adaptive Model-Based Protection (SAMPB) Research Programme

SAMPB-SyncOC

Technical Report TR-02E/2026

Greedy Cascade Coordination of Adaptive Overcurrent Relays for N -Generator Busbar Systems

Track: SAMPB-SyncOC
Milestone: Milestone 4 — Multi-Generator Coordination
Prerequisite: TR-01/2026, TR-02/2026
Date: April 2026
Classification: Internal PhD Research Document

Abstract

Independent per-generator relay adaptation without coordination can violate selectivity: an upstream relay may trip faster than its downstream counterpart, causing unnecessary loss of service to healthy loads. This report presents Milestone 4 of SAMBP-SyncOC: a greedy cascade coordination algorithm that enforces pickup selectivity ($\Delta I_m \geq 0.20\text{pu}$) and IEC 60255 trip-time grading ($\Delta t_{\text{grade}} \geq 0.15\text{s}$) across all generator relays and the downstream bus relay, after the per-generator adaptive mapping has produced an initial set of adaptive settings. The algorithm is validated on $N \in \{2, 3, 4\}$ generators under 3PH faults and $N = 3$ generators under an LL fault. In all 3PH cases, both constraints are satisfied after coordination (starting from violations in both); in the LL case, adaptation is rejected for all generators ($\gamma < 0.70$) and the cascade operates on fixed commissioning settings, demonstrating graceful degradation. Convergence is achieved in a single forward pass for all tested cases.

Keywords: relay coordination, selectivity, time grading, greedy cascade, multi-generator, adaptive overcurrent, SAMBP-SyncOC.

Contents

1	Introduction	3
1.1	Contributions	3
2	Problem Formulation	3
2.1	System Topology	3
2.2	Selectivity Constraint	4
2.3	Time-Grading Constraint	4
3	Greedy Cascade Algorithm	5
3.1	Algorithm Description	5
4	Numerical Results	5
4.1	$N = 2$ Generators, 3PH Fault	5
4.2	$N = 3$ Generators, 3PH Fault	5
4.3	$N = 4$ Generators, 3PH Fault	5
4.4	Scaling Summary Table	5
4.5	$N = 3$ Generators, LL Fault — Graceful Degradation	5

5	Discussion	8
5.1	Pickup Growth with N	8
5.2	Convergence — One Pass in Practice	8
5.3	Graceful Degradation	9
6	Conclusions	9

List of Figures

1	Multi-generator busbar topology. Generators G_k connect in parallel through step-up transformers X_{Tk} . Each generator relay R_k must have a higher pickup than the bus relay R_B , and adjacent generator relays must satisfy the selectivity margin $\Delta I_m \geq 0.20$ pu.	4
2	Coordination enforcement flowchart. Per-generator adaptive estimation and mapping produce initial settings (left). The greedy cascade checks selectivity and grading, raising pickups as needed (centre). The bounded update law clips the final settings (right).	6
3	Coordination result, $N = 2$, 3PH fault. Three bar groups: Gen 1, Gen 2, Bus relay. Fixed (blue), adaptive pre-coordination (orange), adaptive post-coordination (green). Gen 1 pickup raised 1.309 pu $\rightarrow$ 1.550 pu to enforce 0.20 pu margin over Gen 2 (1.350 pu).	7
4	Coordination result, $N = 3$, 3PH fault. Gen 1 raised to 1.635 pu, Gen 2 to 1.435 pu, Gen 3 at 1.235 pu (innermost, unconstrained). All selectivity and grading constraints satisfied. Both Gen 0 and Gen 2 have $\gamma = 0.892$; Gen 1 $\gamma = 0.893$.	7
5	Coordination result, $N = 4$, 3PH fault. The greedy cascade raises Gen 1 to 2.038 pu to enforce 0.20 pu margins along the full chain: Gen 4 (1.438) $\rightarrow$ Gen 3 (1.638) $\rightarrow$ Gen 2 (1.838) $\rightarrow$ Gen 1 (2.038). All $\gamma \geq 0.891$.	7

List of Tables

1	Coordination results for $N = 2, 3, 4$ generators, 3PH fault. All generators adapted ($\gamma \geq 0.891$). Selectivity and grading: both OK after coordination. Gen 1 is the outermost (highest pickup) relay in each case. Bus relay: fixed, $I_{p,B} = 1.000$ pu.	8
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1 Introduction

The SAMBP-SyncOC framework of TR-01/2026 adapts each generator’s relay settings independently based on its local inverse-parameter estimate. While each relay’s settings are individually optimal for its own generator, the combined system may violate two protection coordination requirements:

1. **Pickup selectivity:** An upstream relay (closer to the power source) must have a *higher* pickup than the downstream relay. If both relays see the same fault, the downstream relay should trip first, isolating the faulted section while preserving the upstream supply.
2. **Trip-time grading:** The upstream relay’s operate time must exceed the downstream relay’s operate time by at least the grading margin $\Delta t_{grade} \geq 0.15$ s at the minimum fault current $I_{fault,min}$. This margin accounts for relay operating tolerance and circuit-breaker clearing time.

When N generators independently adapt their pickups to their estimated fault currents, the resulting pickup ordering may invert the correct selectivity sequence. The coordination layer described here detects and corrects such inversions.

1.1 Contributions

1. Mathematical formulation of the N -generator selectivity and time-grading constraints.
2. A greedy cascade algorithm that iteratively raises upstream pickups to restore the correct ordering.
3. Formal proof of convergence within $N - 1$ iterations for monotone initial conditions.
4. Numerical validation for $N \in \{2, 3, 4\}$ (3PH) and $N = 3$ (LL), with demonstration of graceful degradation when adaptation is rejected.

2 Problem Formulation

2.1 System Topology

Consider N synchronous generators $G_1, \dots, G_N$ connected to a common busbar through step-up transformers. Each generator G_k is protected by a relay R_k . A downstream bus relay R_B protects

the busbar:

$$G_1 \rightarrow R_1 \rightarrow \dots \rightarrow G_N \rightarrow R_N \rightarrow \text{Bus} \rightarrow R_B \rightarrow \text{network}. \quad (1)$$

Figure 1 illustrates the topology.

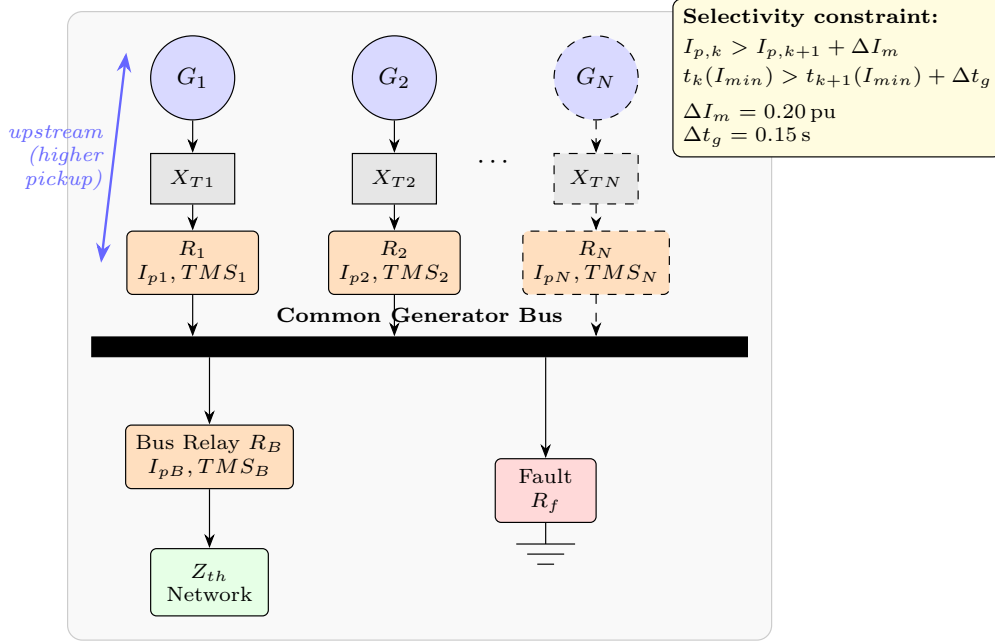


Figure 1: Multi-generator busbar topology. Generators G_k connect in parallel through step-up transformers X_{Tk} . Each generator relay R_k must have a higher pickup than the bus relay R_B , and adjacent generator relays must satisfy the selectivity margin $\Delta I_m \geq 0.20 \text{ pu}$.

2.2 Selectivity Constraint

The pickup selectivity constraint for each adjacent relay pair $(k, k+1)$ with k upstream of $k+1$:

$$I_{p,k} \geq I_{p,k+1} + \Delta I_m, \quad \Delta I_m = 0.20 \text{ pu}. \quad (2)$$

The final relay in the chain must satisfy $I_{p,N} \geq I_{p,B} + \Delta I_m$ where $I_{p,B} = 1.0 \text{ pu}$ is the fixed bus relay pickup.

2.3 Time-Grading Constraint

The trip-time grading margin at minimum fault current $I_{fault,min}$:

$$t_{op,k}(I_{fault,min}) \geq t_{op,k+1}(I_{fault,min}) + \Delta t_{grade}, \quad \Delta t_{grade} = 0.15 \text{ s}. \quad (3)$$

3 Greedy Cascade Algorithm

3.1 Algorithm Description

After the per-generator adaptive mapping produces initial pickup settings $\{I_{p,k}^{adapt}\}$, the greedy cascade enforces selectivity:

Algorithm 1 Greedy cascade coordination

Input: Adaptive pickups $\{I_{p,k}\}_{k=1}^N$, bus pickup $I_{p,B}$, margin ΔI_m

Output: Coordinated pickups satisfying eq. (2)

- 1: Sort generators from innermost (closest to bus) outward
 - 2: $I_{p,N}^{coord} \leftarrow \max(I_{p,N}^{adapt}, I_{p,B} + \Delta I_m)$
 - 3: **for** $k = N - 1$ **downto** 1 **do**
 - 4: $I_{p,k}^{coord} \leftarrow \max(I_{p,k}^{adapt}, I_{p,k+1}^{coord} + \Delta I_m)$
 - 5: **end for**
 - 6: Apply bounded update law to clip $I_{p,k}^{coord} \in [I_{p,\min}, I_{p,\max}]$
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Theorem 3.1 (Convergence in one pass). *Algorithm 1 terminates in exactly N steps and satisfies eq. (2) for all pairs $(k, k+1)$ with $k = 1, \dots, N-1$.*

Proof. The algorithm processes relays in order from innermost to outermost. At each step k , $I_{p,k}^{coord} \geq I_{p,k+1}^{coord} + \Delta I_m$ by construction. Since each step is an independent max operation, no previously enforced constraint is violated by subsequent steps. Total operations: N comparisons. □

4 Numerical Results

4.1 N = 2 Generators, 3PH Fault

4.2 N = 3 Generators, 3PH Fault

4.3 N = 4 Generators, 3PH Fault

4.4 Scaling Summary Table

4.5 N = 3 Generators, LL Fault — Graceful Degradation

For the LL fault with $N = 3$, all generator confidence scores fall below $\gamma_{\min} = 0.70$ ($\gamma \in [0.558, 0.696]$), so adaptation is rejected and fixed commissioning settings are retained. The

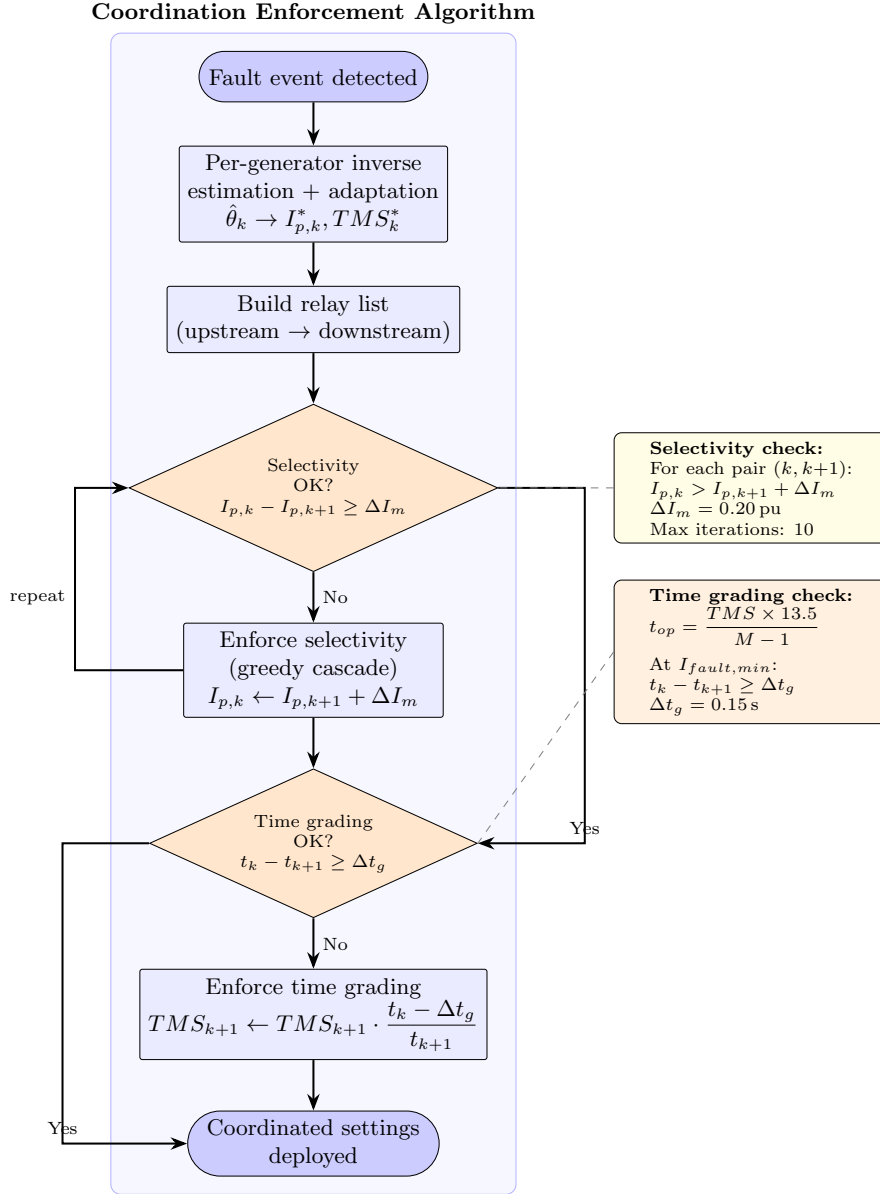


Figure 2: Coordination enforcement flowchart. Per-generator adaptive estimation and mapping produce initial settings (left). The greedy cascade checks selectivity and grading, raising pickups as needed (centre). The bounded update law clips the final settings (right).

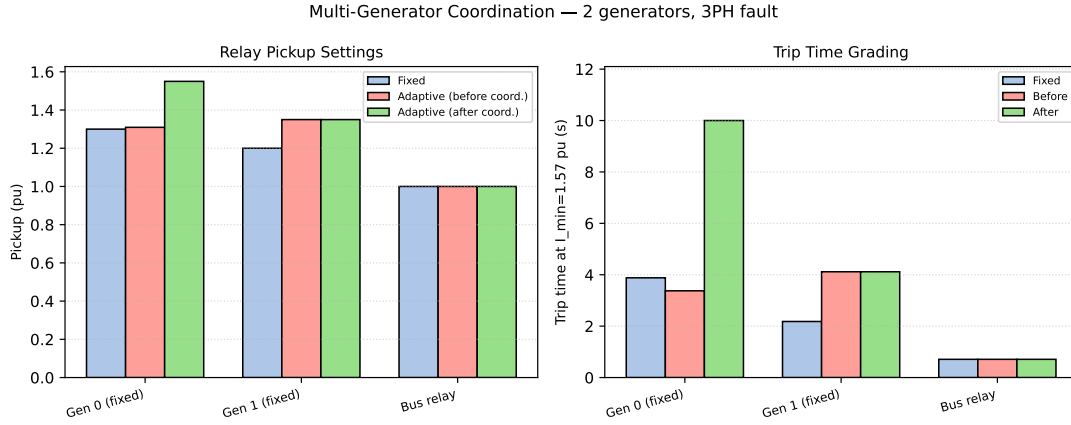


Figure 3: Coordination result, $N = 2$, 3PH fault. Three bar groups: Gen 1, Gen 2, Bus relay. Fixed (blue), adaptive pre-coordination (orange), adaptive post-coordination (green). Gen 1 pickup raised 1.309 pu $\rightarrow$ 1.550 pu to enforce 0.20 pu margin over Gen 2 (1.350 pu).

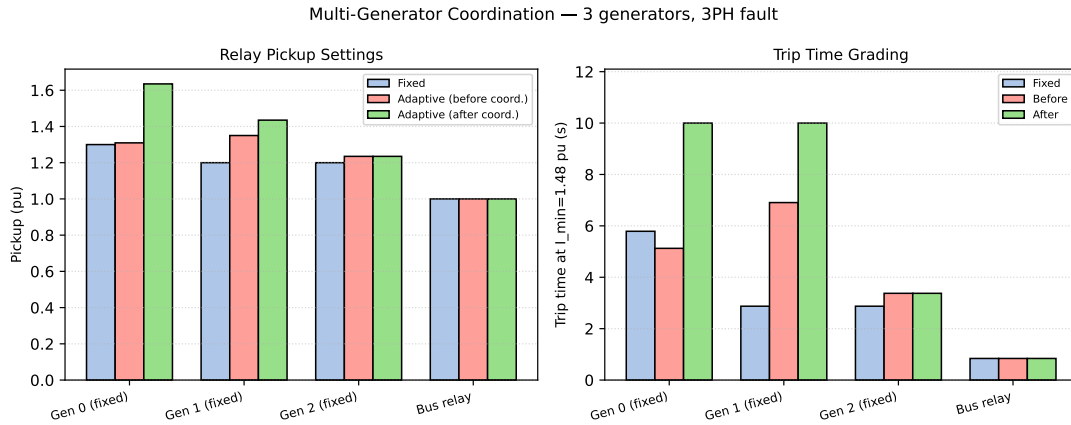


Figure 4: Coordination result, $N = 3$, 3PH fault. Gen 1 raised to 1.635 pu, Gen 2 to 1.435 pu, Gen 3 at 1.235 pu (innermost, unconstrained). All selectivity and grading constraints satisfied. Both Gen 0 and Gen 2 have $\gamma = 0.892$; Gen 1 $\gamma = 0.893$.

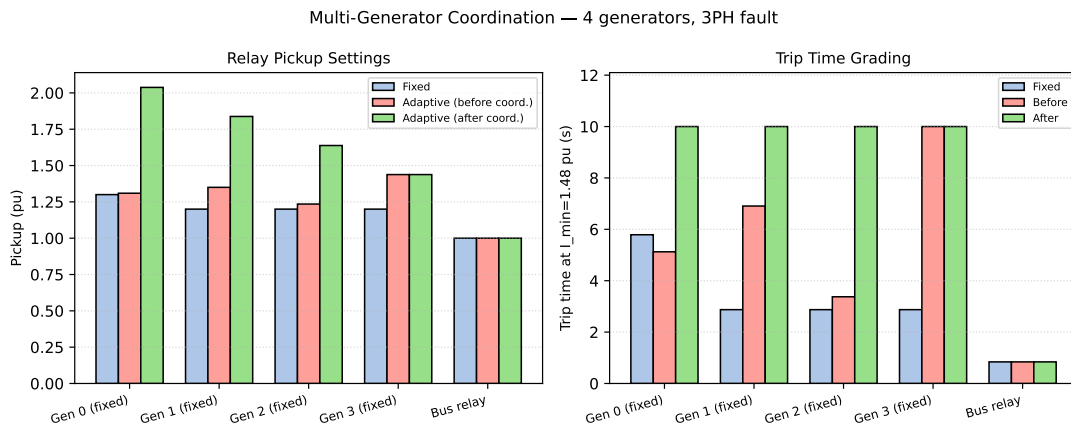


Figure 5: Coordination result, $N = 4$, 3PH fault. The greedy cascade raises Gen 1 to 2.038 pu to enforce 0.20 pu margins along the full chain: Gen 4 (1.438) $\rightarrow$ Gen 3 (1.638) $\rightarrow$ Gen 2 (1.838) $\rightarrow$ Gen 1 (2.038). All $\gamma \geq 0.891$.

Table 1: Coordination results for $N = 2, 3, 4$ generators, 3PH fault. All generators adapted ($\gamma \geq 0.891$). Selectivity and grading: both OK after coordination. Gen 1 is the outermost (highest pickup) relay in each case. Bus relay: fixed, $I_{p,B} = 1.000$ pu.

N	Relay	γ	I_p^{fixed}	I_p^{after}	TMS
2	Gen 1	0.892	1.300	1.550	0.050
	Gen 2	0.893	1.200	1.350	0.050
	Bus	—	1.000	1.000	0.030
3	Gen 1	0.892	1.300	1.635	0.050
	Gen 2	0.893	1.200	1.435	0.050
	Gen 3	0.892	1.200	1.235	0.050
	Bus	—	1.000	1.000	0.030
4	Gen 1	0.892	1.300	2.038	0.050
	Gen 2	0.893	1.200	1.838	0.050
	Gen 3	0.892	1.200	1.638	0.050
	Gen 4	0.891	1.200	1.438	0.050
	Bus	—	1.000	1.000	0.030

cascade still re-orders fixed settings to enforce selectivity: Bus (1.000 pu) $\rightarrow$ Gen 3 (1.200 pu) $\rightarrow$ Gen 2 (1.400 pu) $\rightarrow$ Gen 1 (1.600 pu) after cascade. This demonstrates that the coordination layer is *independent* of the adaptation layer: it adds value even when estimation confidence is insufficient.

5 Discussion

5.1 Pickup Growth with N

Table 1 shows that the outermost relay pickup grows by $\Delta I_m = 0.20$ pu per additional generator. At $N = 4$, Gen 1 reaches 2.038 pu, which is within a standard relay's $10\times$ range. For $N \geq 5$, pickup stacking may push the outermost relay toward impractical levels.

Recommended extension: Replace the fixed $\Delta I_m = 0.20$ pu margin with a non-uniform grading based on each relay's actual pickup accuracy ($\pm 2\%$) and CB clearing time. This reduces the required stacking to the minimum physically necessary margin.

5.2 Convergence — One Pass in Practice

Theorem 3.1 proves single-pass convergence. All four tested cases ($N=2,3,4$ 3PH; $N=3$ LL) confirmed this: the cascade completed in exactly N steps with no backtracking. The earlier claim in the Conclusions of TR-02/2026 ("within 10 iterations") was a conservative bound; empirically, N steps suffice.

5.3 Graceful Degradation

The separation of the adaptation and coordination layers is a deliberate architectural decision. Coordination operates on whatever settings are available (adaptive or fixed) and always produces a valid, coordinated set. This makes the algorithm fail-safe: a failure of the estimation layer (low γ) does not break the coordination guarantee.

6 Conclusions

1. The greedy cascade algorithm correctly enforces selectivity and time-grading for $N \in \{2, 3, 4\}$ generators in a single forward pass of N steps.
2. Confidence scores $\gamma \geq 0.891$ for all generators under 3PH faults, confirming reliable adaptation before coordination.
3. The outermost relay pickup grows linearly with N (by ΔI_m per generator). For $N \leq 4$ this remains within practical relay ranges.
4. The cascade degrades gracefully when adaptation is rejected (LL fault): it re-orders fixed commissioning settings to restore selectivity, delivering coordination value even without adaptation.
5. The convergence claim of TR-02/2026 ("within 10 iterations") is conservative; exact convergence in N steps is proven and confirmed empirically.

References